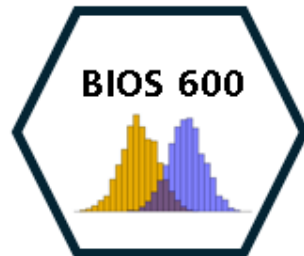


Lecture 12: ANOVA

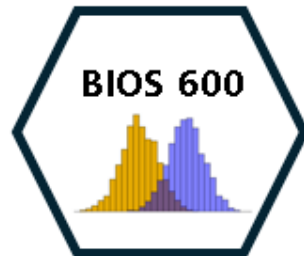
BIOS 600 - Fall 2025

UNC - Chapel Hill

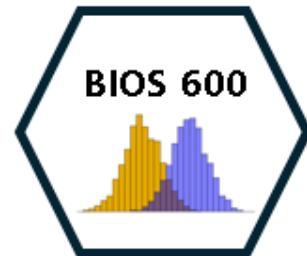


Announcements

- ▶ Great to hear that the pre-recorded video is helpful! We will keep doing that.
- ▶ Looking at your responses re: tutoring times and will discuss with the tutoring team to see if they can offer different times.
- ▶ HW 5 due tonight at 11:59pm.

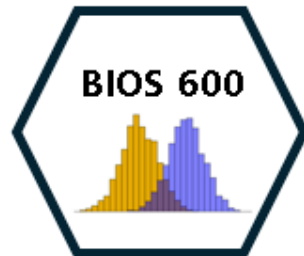


Overview

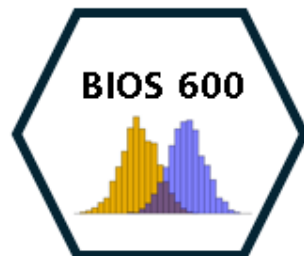


Very optional reading

- ▶ P&G: Chapter 12
- ▶ OI: 7.5

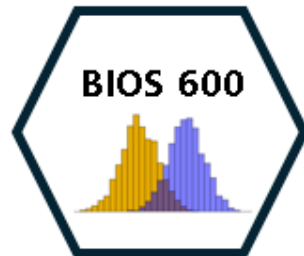


Puppies

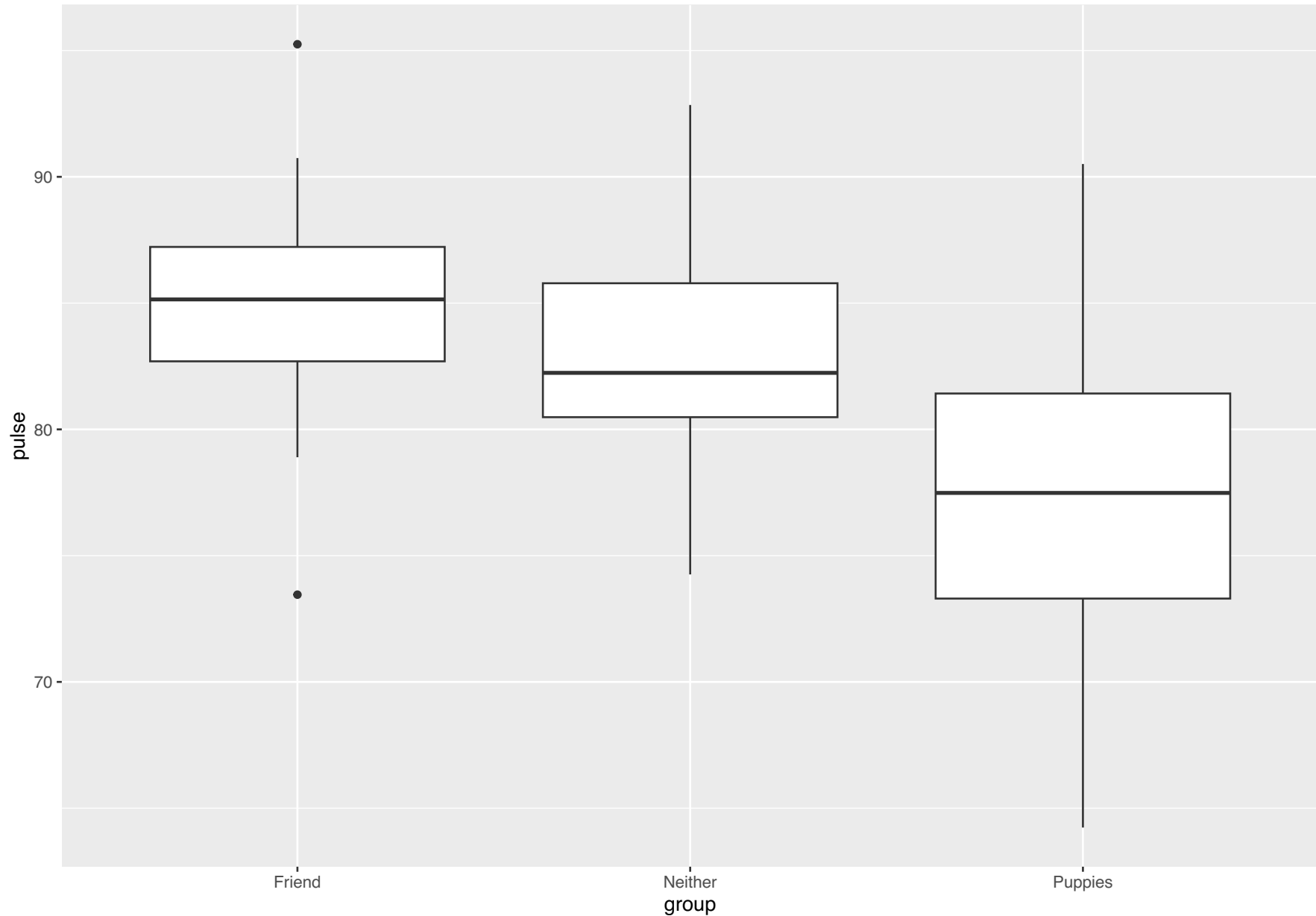


Experiment: Pets and Stress Level

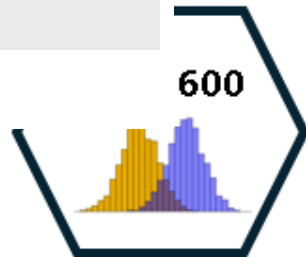
- ▶ For today's topic, we will consider an experiment where heart rates were measured (as a proxy for stress level) for a group of $n = 90$ students after 10 minutes of being under certain conditions.
- ▶ **The 3 Conditions/Groups:** Of the 90 total students, 30 students spent time holding puppies, 30 students sat alone, and 30 students chatted with a friend.
- ▶ After 10 minutes, each student's **heart rate** was measured.
- ▶ **Research Question:** Is there a group (puppies, friend, alone/neither) associated with lower heart rate compared to the others?
- ▶ Let's take a look at the data on the next slide.



Distribution of heart rate data



How do we compare across three groups? Which groups are different?



Example: pets and stress

We are interested in testing

μ_P : mean pulse in puppy group

μ_F : " " friend group

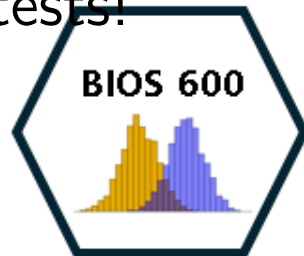
μ_N : " " neither group

$$H_0 : \mu_P = \mu_F = \mu_N$$

against the alternative that at least one mean is different from the others.

- ▶ One way to do this would be to use t-tests on all possible pairs of tests (here there are just three - how many pairs of tests would this be?). However, if we have more groups, this becomes quite complicated.
- ▶ For example, with 10 groups we need to do $\binom{10}{2} = 45$ tests!

10 choose 2



Review: Error rates

💡 Tip

In groups, fill in the blanks:

- ▶ Suppose we test the null hypothesis $H_0: \mu = \mu_0$. We could potentially make two types of errors:

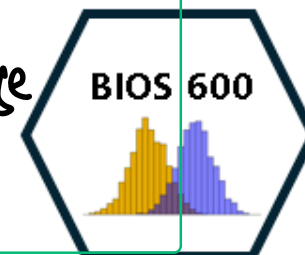
Truth	$\mu = \mu_0$	$\mu \neq \mu_0$
Fail to reject H_0	correct decision	Type II
Reject H_0	Type I	correct decision

0.95

$\alpha = .05$

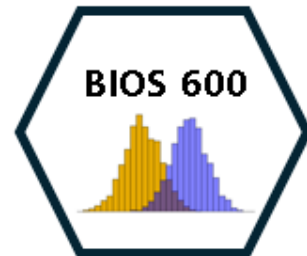
How would you explain the following in words?

- ▶ **Type I Error:** reject H_0 when H_0 actually true (false positive)
- ▶ **Type II Error:** fail to reject H_0 when H_0 false

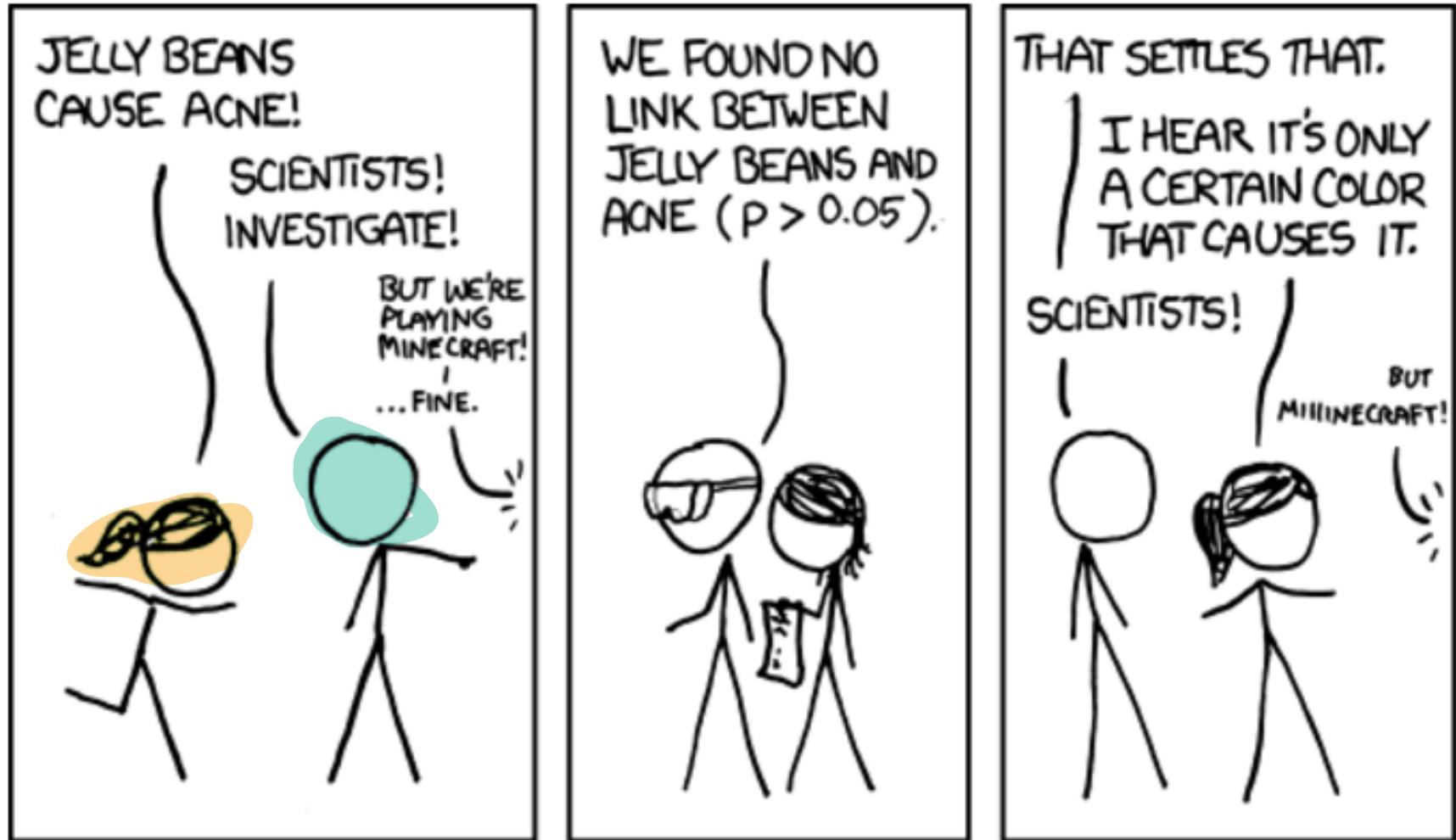


Multiple comparisons

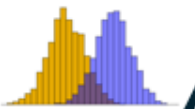
- ▶ Now that we've reviewed Type I and Type II errors, we need to discuss **multiple comparisons** - an important concept in statistics.
- ▶ Back to our pet example. What if we had 10 groups?
- ▶ In addition to being time-consuming, carrying out multiple tests can lead to **inflated Type I error rates** which puts into question the validity of a given study if these **multiple comparisons** are not accounted for. ✖



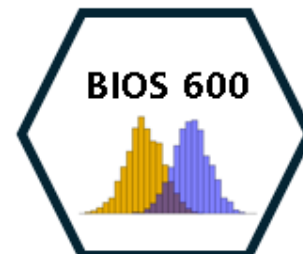
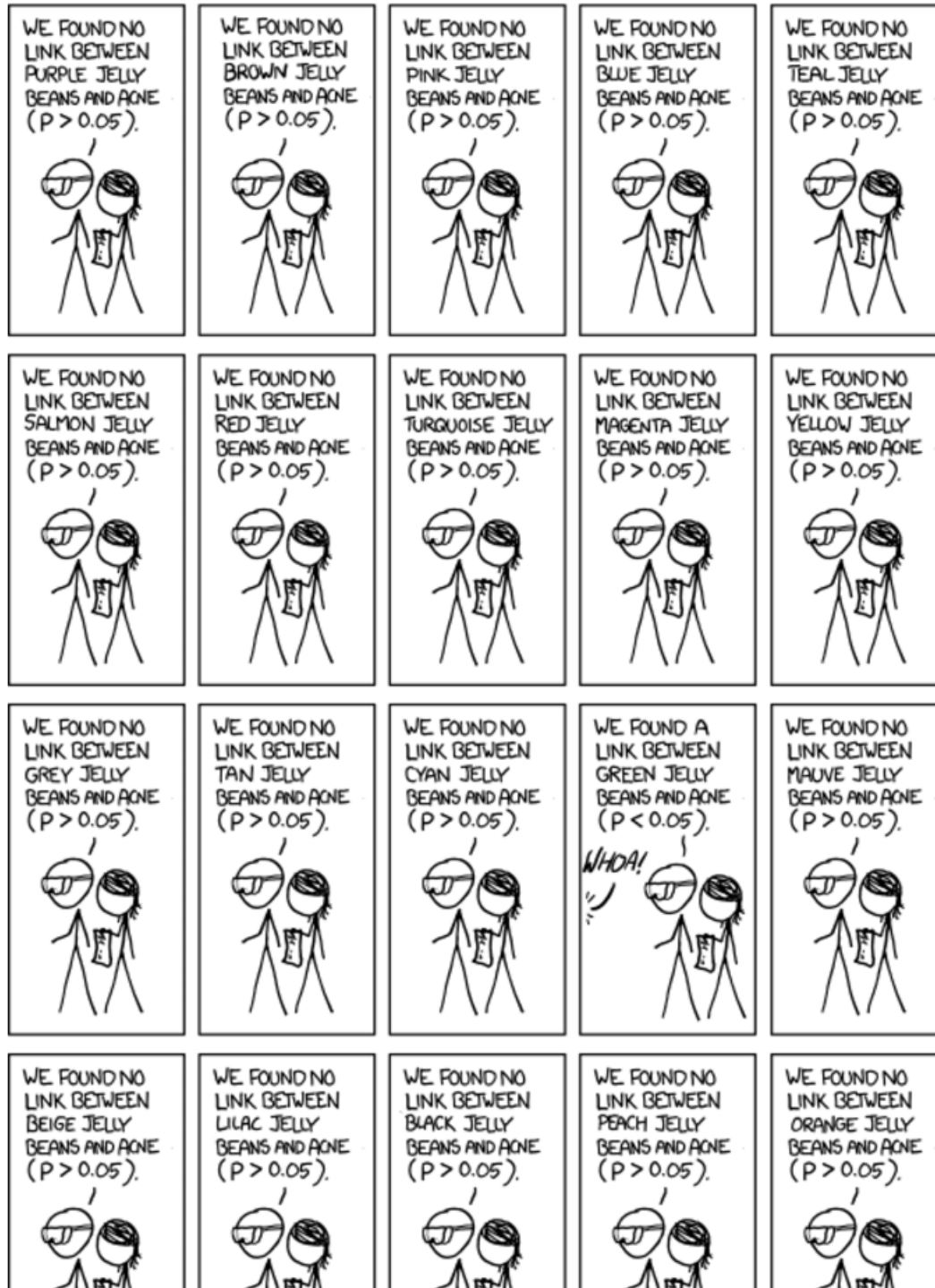
Multiple comparisons in action: overall test



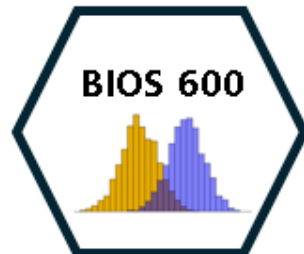
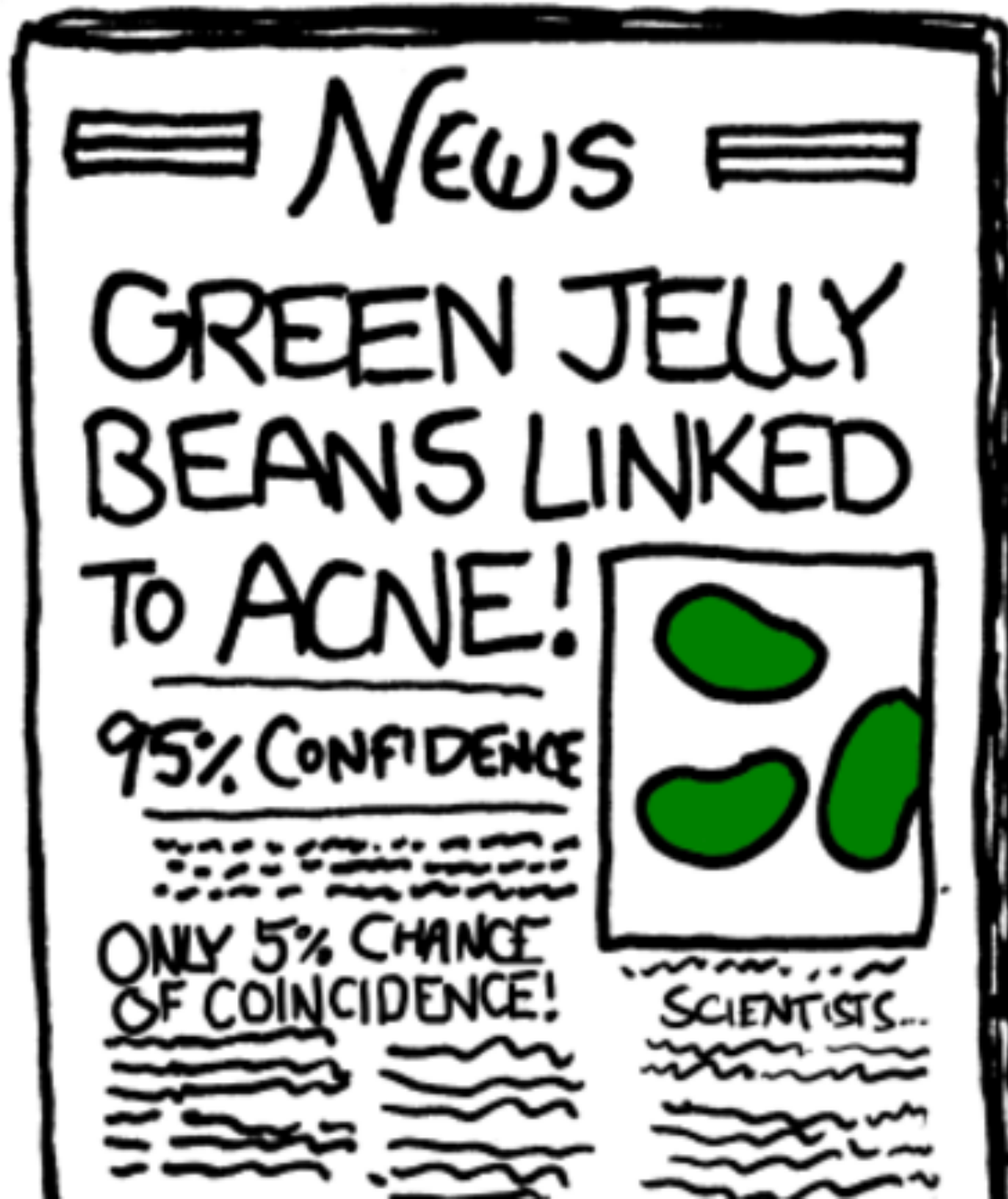
BIOS 600



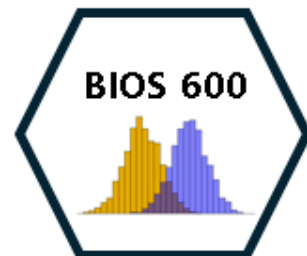
Extra tests...whoa!



Green jelly beans!



So what went wrong?



Multiple comparisons

Consider our pets / stress example, where we had three pairwise comparisons.

► Suppose all means are truly equal and we conduct all three pairwise tests

1. $H_0: \mu_P = \mu_F$ vs. $H_A: \mu_P \neq \mu_F$

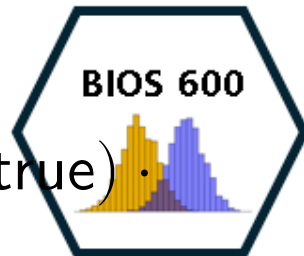
2. $H_0: \mu_P = \mu_N$ vs. $H_A: \mu_P \neq \mu_N$

3. $H_0: \mu_N = \mu_F$ vs. $H_A: \mu_N \neq \mu_F$

► Suppose also the tests are independent and done at a 0.05 significance level

► Then the probability we fail to reject **all three tests if the null hypothesis were true** (i.e. the **correct decision**), assuming the tests are independent, is:

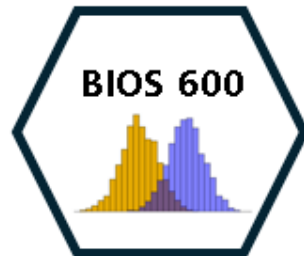
$$P(\text{fail to reject Test 1} | H_0 \text{ true}) \cdot P(\text{fail to reject Test 2} | H_0 \text{ true}) \cdot P(\text{fail to reject Test 3} | H_0 \text{ true})$$



Multiple comparisons (continued)

$$= (.95) \cdot (.95) \cdot (.95)$$

- ▶ Assuming the tests are independent, this probability is $(1 - 0.05)^3 = 0.95^3 = 0.857$, and so the probability of rejecting at least one of the three null hypotheses, called the **family-wise error rate**, is $1 - 0.857 = 0.143 > 0.05$
- ▶ If we had 10 groups instead, we would need 45 pairwise tests. With 45 tests, the probability of rejecting at least 1 of them (incorrectly!) is over 90%!



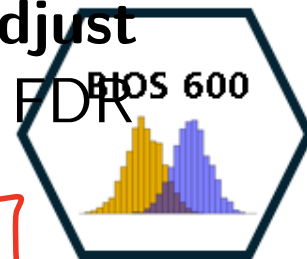
Intuitive Takeaway: Why Multiple Comparisons Matter

- ▶ Each hypothesis test carries a small chance (e.g., 5%) of being a **false positive**.
 - ▶ Like flipping a coin: occasionally, you'll get "heads" just by chance.
- ▶ When we run **many tests**, those small chances add up.
 - ▶ With only 3 tests, there's already a 14% chance of at least one false alarm. With 45 tests, that chance exceeds 90%!
- ▶ In practice: Even if nothing real is going on, we're **almost guaranteed** to find something "statistically significant." This can lead to **misleading conclusions** — especially in large studies testing many variables.
- ▶ **Takeaway:** The more tests we do, the more we must **adjust for multiple comparisons** (e.g., Bonferroni correction, FDR (false discovery) control) to avoid fooling ourselves.

10 tests

$\alpha/10$ $\alpha = .05$

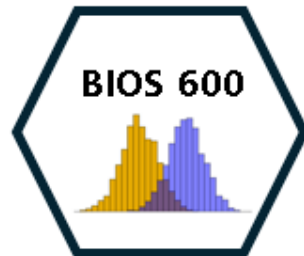
$.05/10 = .005$



Multiple comparisons

group 1 group 2

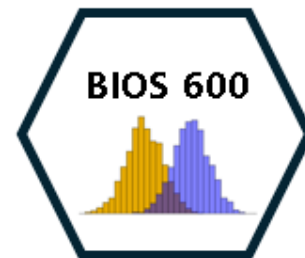
- ▶ **ANOVA** extends the t-test and is one way to control the overall Type I error rate at a fixed level α , if we only test pairwise differences when the overall ANOVA test is rejected
- ▶ ANOVA stands for **analysis of variance**. We use ANOVA when we want to compare more than two groups.



ANOVA null hypothesis

In ANOVA, we typically follow this testing procedure:

1. First, we conduct an **overall** test of the null hypothesis that the means of all of the groups are equal.
2. If this is rejected, then we **step down** to see which means are different from each other. A **multiple comparisons** correction is sometimes used for these pairwise comparisons of means.
3. If we fail to reject the null hypothesis, then no further testing should be done.

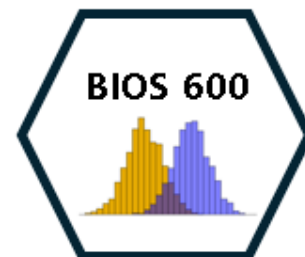


ANOVA alternative hypothesis

For ANOVA with three groups, our null hypothesis is $H_0 : \mu_P = \mu_F = \mu_N$. What could happen under the alternative?

- ▶ $\mu_P \neq \mu_F \neq \mu_N$
- ▶ $\mu_P = \mu_F$, but μ_N is different
- ▶ $\mu_P = \mu_N$, but μ_F is different
- ▶ $\mu_F = \mu_N$, but μ_P is different

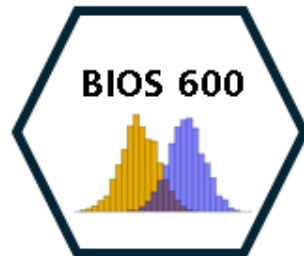
The alternative hypothesis for ANOVA is that at least one of the means is different from the others.



Assumptions of ANOVA

1. Outcomes within groups are normally distributed
2. Homoscedastic variance (the within-group variance is the same for all groups)
3. Samples are independent

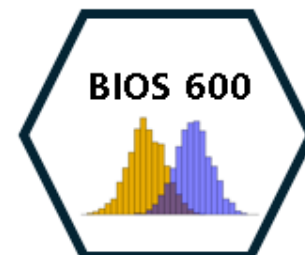
If these assumptions are violated, then results from ANOVA may not be valid. We will discuss some alternatives later in the course.



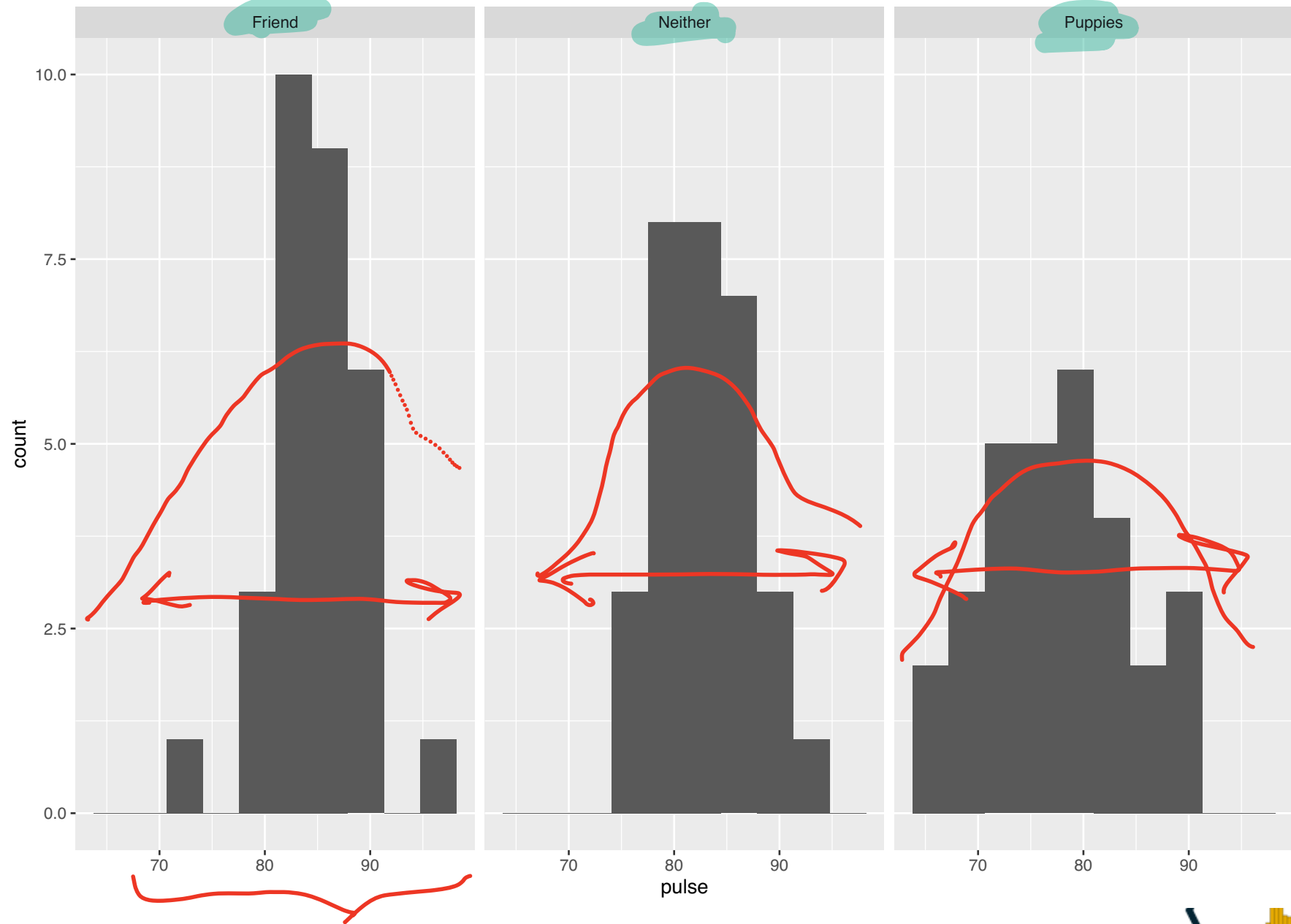
Validity of ANOVA for pet data

<u>group</u>	<u>pulse</u>
puppy	—
puppy	—
⋮	
friend	—

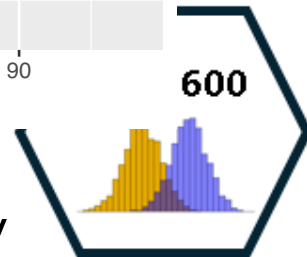
```
pulse_data |>  
  ggplot(mapping = aes(x = pulse)) +  
  geom_histogram(bins=10) +  
  facet_wrap(~group)
```



Validity of ANOVA for pet data



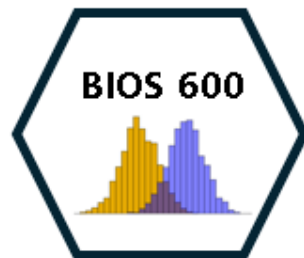
Variances appear to be similar, but normality may be slightly questionable. For now, let's proceed despite this.

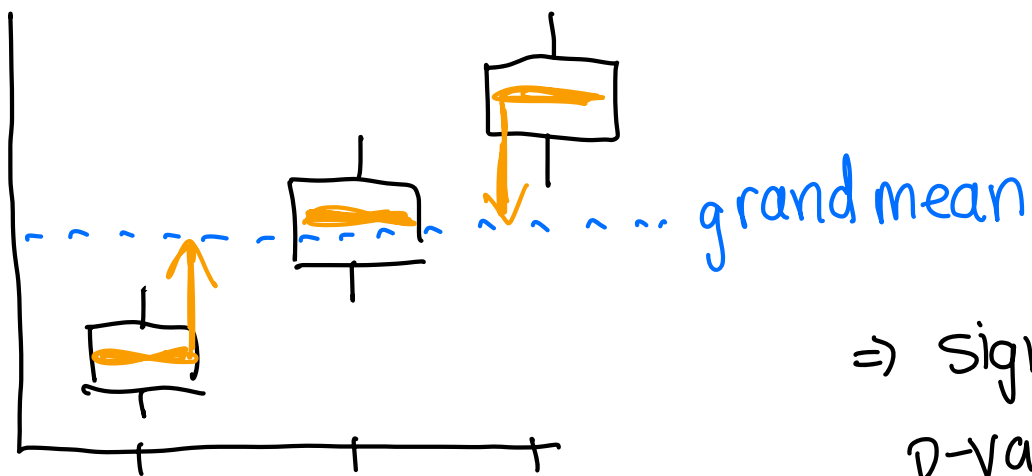


Why analyze variance when we want means?

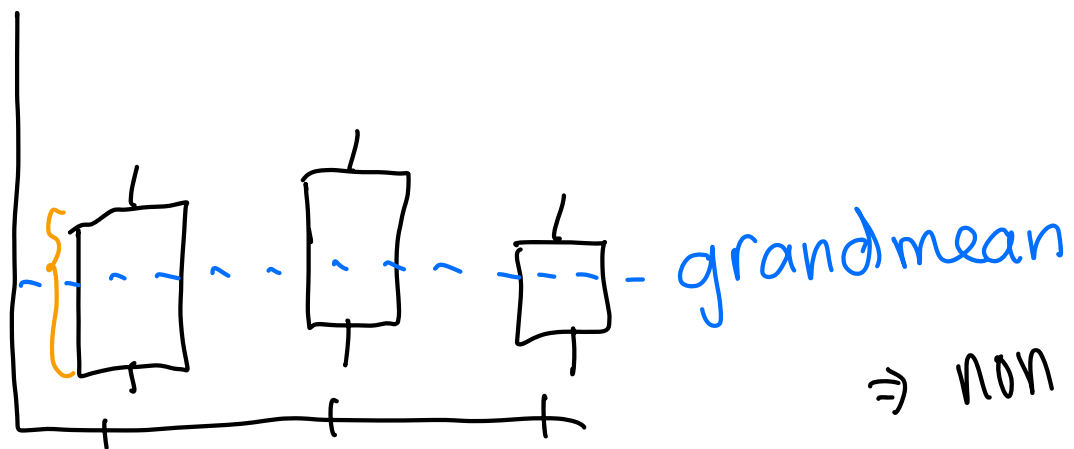
- ▶ Remember, ANOVA stands for analysis of variance.
- ▶ What does variance have to do with our null hypothesis, which is about equality of means (say, $H_0 : \mu_1 = \mu_2 = \dots = \mu_K$)?

F-test: If the **group-specific means** vary around the **overall grand mean** more than the individual observations vary around their group-specific sample means, then we have evidence that the corresponding population means are in fact different. [Drawing]





⇒ significant
p-val in
ANOVA



⇒ non-significant
p-value

F-test

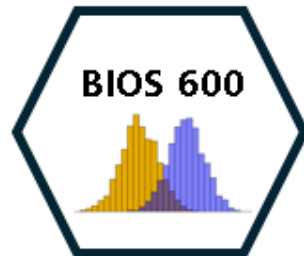
$$F_{K-1, n-K}$$

How do we formally compare these variances? Consider the F statistic given by

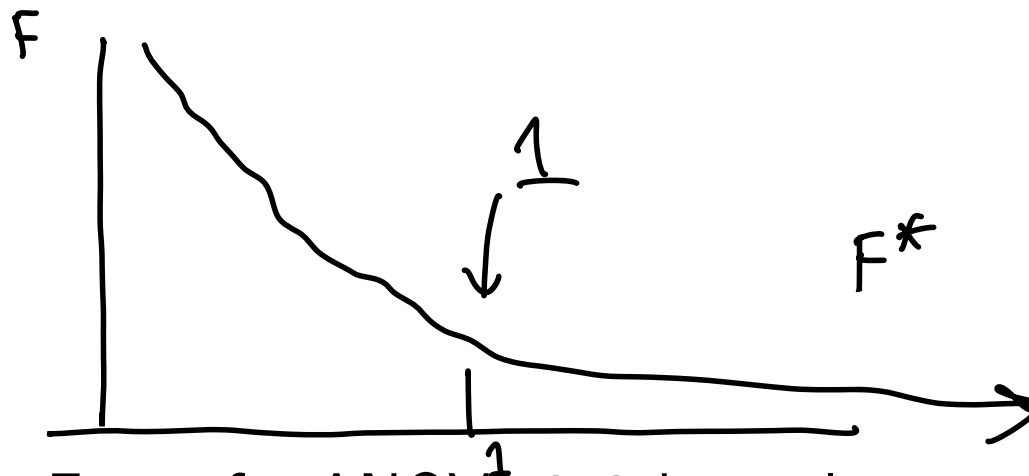
$$F = \frac{s_{between}^2}{s_{within}^2}$$

$K = \# \text{ of groups}$
 $K-1$
 $n-K$

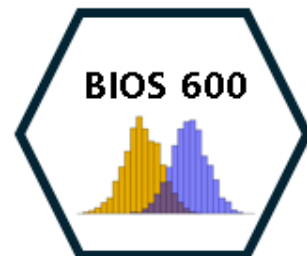
which if H_0 is true, has an F distribution with $K-1$ **numerator** degrees of freedom and $n - K$ **denominator** degrees of freedom, where K is the number of groups and n is the total number of observations.



F-test



- ▶ The F-test for ANOVA is inherently one-tailed, rejecting H_0 only if F is considerably larger than one.
- ▶ Importantly, this does not mean we have a one-sided alternative; we just look at one tail of the F distribution to get the p-value.
- ▶ If there are only two groups, then the F-test gives the same result as the independent-samples t-test.



Output

Outcome measurement
↓

```
# anova in R:  
summary(aov(pulse ~ group, data = pulse_data))
```

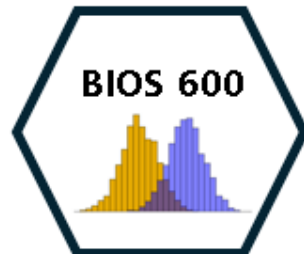
	Df	Sum Sq	Mean Sq	F value	Pr(>F)
*group	2	877.7	438.8	15.77	1.44e-06 ***
Residuals	87	2421.7	27.8		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 '

💡 Which rows correspond to the between-group and within-group variances here, respectively?

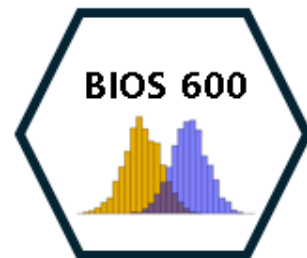


$$\frac{877.7/2}{2421.7/87}$$



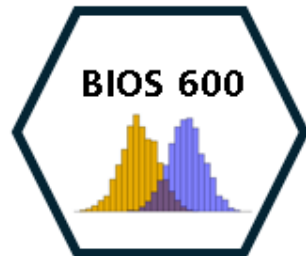
F-test for pet data $\overset{1}{15} = 15.77$

- ▶ Note that $F = s_B^2 / s_W^2 = \frac{(877.7/2)}{(2421.7/87)} = 15.77$, with ndf = $3 - 1 = 2$ and ddf $90 - 3 = 87$.
- ▶ This corresponds to a p-value $< .0001$.
- ▶ **Conclusion:** At $\alpha = 0.05$, we reject the null hypothesis. There is sufficient evidence that at least one of the three groups comes from a population with a different mean than the others.
- ▶ Which groups might be different?



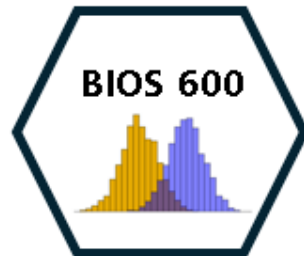
Application Exercise

- ▶ On Canvas, download today's Application Exercise template.
- ▶ The application Exercise will be due **Monday at 11:59pm**.



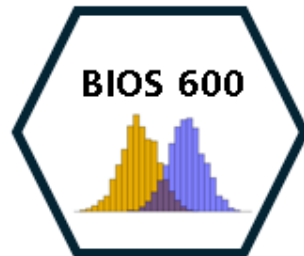
Bonferroni correction

- ▶ As we showed earlier, conducting multiple tests on a data set increases the **family-wise error rate**.
- ▶ One very conservative way to ensure this is not the case is to simply divide α by the number of tests to be done and to use that as the significance level.
- ▶ This procedure is called the **Bonferroni correction**.



Bonferroni correction

- ▶ For example: for two tests, to preserve an overall 0.05 type I error rate, the Bonferroni correction would use $\alpha/2 = 0.025$ as the significance level for each individual test instead of 0.05.
- ▶ Bonferroni is a conservative correction, making it harder to reject the null hypothesis, but it is a safe bet in controlling the Type I error rate.



Pets and stress: group differences

- ▶ We can compare the groups using a Bonferroni correction (here we have three tests, so the significance level for each test is $\alpha/3$).
- ▶ The raw (uncorrected) p-values for the t-test comparing friend vs. pet was < 0.0001 ; for friend vs. neither was 0.021, and for pet vs. neither was 0.009.

💡 In your small groups: What might we conclude?

